

Chapter 4

Word Problems II: Rates, Ratios, and Changing Quantities

A word problem is not mainly a test of arithmetic. It is a small exercise in *mathematical modeling*: we decide what quantities matter, describe how they are related, and then use mathematics to answer the question.

In this chapter we study several situations in which quantities change: proportions, mixtures, motion, work, and clocks. The formulas are useful, but the more important goal is to learn how to *derive* them. A future scientist or mathematician should not only know a formula, but should also be able to explain where it comes from, when it applies, and what new questions it suggests.

4.1 Start with Relationships, Not Formulas

Before writing an equation, ask four questions.

1. What is unknown?
2. Which quantities are changing, and which are fixed?
3. What relationship stays true?
4. What units should the answer have?

Example 4.1. A number is multiplied by 4 and then 12 is subtracted. The result equals twice the result obtained by first subtracting 12 from the number and then multiplying by 8. Find the number.

Solution. Let the number be n . Translating the two descriptions gives

$$4n - 12 = 2\{8(n - 12)\}.$$

Thus

$$4n - 12 = 16n - 192,$$

so $180 = 12n$ and

$$n = 15.$$

Notice that the important step was not solving the equation. It was translating the two procedures correctly.

Ask a new question. Replace 4, 12, 2, and 8 by variables. Under what conditions does the resulting equation have exactly one solution?

4.2 Ratios and Proportional Change

If two quantities A and B remain in a fixed ratio $p : q$, then

$$\frac{A}{B} = \frac{p}{q}.$$

This simple idea appears in scale drawings, recipes, unit conversions, mixtures, and many other settings.

Example 4.2. A two-inch interval on a ruler, from 15 to 17, is divided so that the left part and right part are in the ratio $3 : 5$. Where is the dividing point?

Solution. Let its coordinate be x . Then

$$\frac{x - 15}{17 - x} = \frac{3}{5}.$$

Hence

$$5(x - 15) = 3(17 - x),$$

so $8x = 126$ and

$$x = 15\frac{3}{4}.$$

There is also a faster way to see the structure. The entire length 2 is split into $3 + 5 = 8$ equal ratio-parts. The left piece has length $2(3/8) = 3/4$, giving the same answer.

Discovery. Suppose an interval from a to b is divided in the ratio $p : q$. Derive the coordinate x of the dividing point.

Since

$$\frac{x - a}{b - x} = \frac{p}{q},$$

we obtain

$$q(x - a) = p(b - x),$$

and therefore

$$x = \frac{qa + pb}{p + q}.$$

This formula is worth deriving, not memorizing.

Example 4.3. A laboratory mixture contains 4 ounces of a dissolved substance and 12 ounces of water. Water evaporates, but the dissolved substance does not. How many ounces of water must evaporate so that the dissolved substance is 40% of the final mixture?

Solution. Let x ounces of water evaporate. The amount of dissolved substance remains 4 ounces, while the total amount becomes $16 - x$. Thus

$$\frac{4}{16 - x} = \frac{40}{100} = \frac{2}{5}.$$

Solving gives $20 = 32 - 2x$, so

$$x = 6.$$

The key modeling idea is that evaporation changes the amount of water but not the amount of dissolved substance.

4.3 Motion: Distance Equals Rate Times Time

For constant speed,

$$d = rt.$$

Rather than treating this as a formula to memorize, remember its meaning: rate tells us how much distance is covered in one unit of time.

Catch-up problems

Suppose a slower traveler has a head start of h hours. The slower speed is v and the faster speed is u , where $u > v$. If the faster traveler needs t hours to catch up, then both have traveled the same distance:

$$ut = v(t + h).$$

Therefore

$$(u - v)t = vh,$$

so

$$t = \frac{vh}{u - v}.$$

This is the familiar idea

$$\text{catch-up time} = \frac{\text{head-start distance}}{\text{relative speed}}.$$

Example 4.4. Maya bicycles at 9 miles per hour. Twenty minutes later, Leo starts from the same point along the same route at 12 miles per hour. How long after Leo starts will he catch Maya?

Solution. Maya's head start is

$$9 \left(\frac{20}{60} \right) = 3 \text{ miles.}$$

Leo gains at $12 - 9 = 3$ miles per hour. Thus the catch-up time is

$$\frac{3}{3} = 1 \text{ hour.}$$

Ask a new question. If Leo rode 13 miles per hour instead, would the catch-up time be cut in half? Predict first, then calculate.

Travel with wind or current

If an object moves at speed v in still air or still water and the wind or current has speed c , then

$$\text{speed with the current} = v + c,$$

$$\text{speed against the current} = v - c.$$

Example 4.5. A boat moves at 10 miles per hour in still water. It travels upstream for 30 minutes and returns the same distance downstream in 20 minutes. Find the speed of the current.

Solution. Let the current speed be c . Equal distances give

$$(10 - c)\frac{1}{2} = (10 + c)\frac{1}{3}.$$

Multiplying by 6,

$$3(10 - c) = 2(10 + c).$$

Thus $30 - 3c = 20 + 2c$, so

$$c = 2 \text{ miles per hour.}$$

Discovery. Suppose the upstream time is t_1 , the downstream time is t_2 , and the still-water speed is v . From

$$(v - c)t_1 = (v + c)t_2$$

derive

$$c = v \frac{t_1 - t_2}{t_1 + t_2}.$$

Before accepting the formula, ask: why should $t_1 > t_2$ when $c > 0$?

Example 4.6. Two towns are 180 miles apart. Two cars start at the same time and drive toward each other at 40 and 50 miles per hour. How far from the first town do they meet?

Solution. Their closing speed is $40 + 50 = 90$ miles per hour, so they meet after

$$\frac{180}{90} = 2 \text{ hours.}$$

The first car travels

$$40(2) = 80 \text{ miles.}$$

4.4 Work Problems: Add Rates, Not Times

If one machine completes one job in a hours, then in one hour it completes $1/a$ of the job. Its work rate is therefore

$$\frac{1}{a} \text{ job per hour.}$$

If two machines work independently and their rates add, then

$$\frac{1}{T} = \frac{1}{a} + \frac{1}{b},$$

where T is their combined time.

Example 4.7. Pump A empties a tank in 6 hours. Pumps A and B together empty it in 4 hours. How long would Pump B take alone?

Solution. Pump A's rate is $1/6$ tank per hour and the combined rate is $1/4$. Hence Pump B's rate is

$$\frac{1}{4} - \frac{1}{6} = \frac{1}{12}.$$

Therefore Pump B alone needs

12 hours.

Discovery. If one worker needs a hours and the two workers together need T hours, derive the time b needed by the second worker:

$$\frac{1}{b} = \frac{1}{T} - \frac{1}{a} = \frac{a - T}{aT},$$

so

$$b = \frac{aT}{a - T}.$$

What restriction on T is necessary? Why does it make sense in the real situation?

Example 4.8. One printer can finish a large job in 8 minutes and a second printer can finish it in 12 minutes. How long do they need together?

Solution.

$$\frac{1}{T} = \frac{1}{8} + \frac{1}{12} = \frac{5}{24},$$

so

$$T = \frac{24}{5} = 4.8 \text{ minutes.}$$

Notice that the combined time is less than either individual time. This is a useful reasonableness check.

4.5 Clocks: Derive the Formula

Clock problems are a good example of why deriving a formula is more valuable than memorizing one.

At $h : m$, measured from 12 o'clock,

$$\text{minute-hand angle} = 6m \text{ degrees,}$$

because the minute hand turns 360 degrees in 60 minutes.

The hour hand turns 30 degrees each hour and also moves $1/2$ degree each minute. Thus

$$\text{hour-hand angle} = 30h + \frac{m}{2}.$$

Therefore the difference between the two directions is

$$\left| 30h - \frac{11}{2}m \right|.$$

If this difference is greater than 180 degrees, the smaller angle is 360 degrees minus the difference.

Example 4.9. Find the smaller angle between the hands at 5:15.

Solution.

$$\left| 30(5) - \frac{11}{2}(15) \right| = |150 - 82.5| = 67.5 \text{ degrees.}$$

Example 4.10. Between 2:00 and 3:00, when are the hands 20 degrees apart?

Solution. Put $h = 2$ and let m be the number of minutes after 2:00. Then

$$\left| 60 - \frac{11}{2}m \right| = 20.$$

Thus

$$60 - \frac{11}{2}m = 20 \quad \text{or} \quad 60 - \frac{11}{2}m = -20.$$

Hence

$$m = \frac{80}{11} \quad \text{or} \quad m = \frac{160}{11}.$$

So there are two times, approximately 2:07:16 and 2:14:33.

Ask a new question. Between 2:00 and 3:00, for which angles are there two solutions? For which angles is there only one? A single solved problem can lead to an entire family of questions.

4.6 Changing Units and Inventing New Units

Proportional reasoning also allows us to create or convert systems of units.

Example 4.11. Imagine that one day is divided into 10 new hours, each new hour into 50 new minutes, and each new minute into 60 new seconds. What new time corresponds to 2:00 p.m.?

Solution. At 2:00 p.m., $14/24$ of a day has passed. Therefore the number of new hours is

$$10 \left(\frac{14}{24} \right) = \frac{35}{6} = 5 + \frac{5}{6}.$$

The fractional part gives

$$\frac{5}{6}(50) = 41 + \frac{2}{3}$$

new minutes, and

$$\frac{2}{3}(60) = 40$$

new seconds. Thus the new time is

$$5 \text{ hours } 41 \text{ minutes } 40 \text{ seconds.}$$

What Have We Discovered?

The examples in this chapter look different, but a small number of ideas connect them.

1. Translate relationships before doing arithmetic.

2. Ratios describe how quantities compare; proportions turn those comparisons into equations.
3. In motion problems, compare distances and use relative speed when it simplifies the situation.
4. In work problems, add *rates*, not completion times.
5. A useful formula should be derived from simpler facts whenever possible.
6. Units and reasonableness checks can reveal modeling or arithmetic errors.
7. After solving one problem, ask what happens if the numbers are replaced by variables or one assumption is changed.

Do not stop when you get an answer. Ask what the answer teaches you.

Exercises

Problems 1–10 are Core Exercises and contain the essential work for this chapter. Problems 11–20 are Challenge Problems at approximately AMC 8 and MATHCOUNTS level. They are optional for students whose main goal is a solid mathematical foundation, but are strongly recommended for students preparing for competitions. All exercises below have been newly written for this book.

Part A. Core Exercises

1. A 30-inch ribbon is cut into two pieces in the ratio $2 : 3$. Find the length of each piece.
2. A point P lies between 4 and 19 on a number line. If the ratio of $P - 4$ to $19 - P$ is $2 : 3$, find the coordinate of P .
3. A drink contains 6 ounces of syrup and 18 ounces of water. How much water must evaporate so that syrup becomes 40% of the remaining mixture?
4. Nina walks at 3 miles per hour. Forty minutes later, Omar starts from the same point at 5 miles per hour. How many hours after Omar starts will he catch Nina?
5. Two cyclists start 126 miles apart and ride toward each other at 16 and 20 miles per hour. How long does it take them to meet?
6. A boat travels at 12 miles per hour in still water. The current is 3 miles per hour. How long does it take the boat to travel 30 miles upstream?
7. One pipe fills a tank in 10 hours and another fills it in 15 hours. How long will they take if both are used together?
8. Machine A completes a job in 9 hours. Machines A and B together complete the job in 6 hours. How long would Machine B take alone?
9. Find the smaller angle between the hands of a clock at 4:20.
10. A recipe uses flour, oats, and nuts in the ratio $5 : 3 : 2$. If 4 pounds of the mixture are prepared, how many pounds are oats?

Part B. Challenge Problems

11. A point P divides the interval from a to b internally in the ratio $2 : 5$. If $P = 18$ and $a = 8$, find b .
12. A 20% salt solution has 5 ounces of pure water evaporated from it. The resulting solution is 25% salt. How many ounces were in the original solution?
13. Two runners start at the same point on a circular track and run in the same direction. One runs at 7 meters per second and the other at 5 meters per second. The track is 300 meters long. How many seconds after the start will the faster runner first lap the slower runner?

14. A train 180 meters long passes a pole in 12 seconds. At the same constant speed, how many seconds will it take the entire train to pass through a tunnel 420 meters long?
15. A plane flies 600 miles with a steady wind and returns the same 600 miles against the wind. Its speed in still air is 250 miles per hour, and the total flying time is 5 hours. Find the wind speed.
16. Pipe A fills a tank in 6 hours, Pipe B fills it in 8 hours, and a drain can empty a full tank in 24 hours. If all three are open, how long will it take to fill an empty tank?
17. At what time between 3:00 and 4:00 do the hands of a clock first form a right angle? Give the number of minutes after 3:00 as a fraction.
18. A car travels the first half of a 120-mile trip at 40 miles per hour and the second half at 60 miles per hour. What is its average speed for the entire trip?
19. Worker A can complete a job in 12 hours. A and B work together for 3 hours, after which A leaves. B finishes the remaining job in 6 more hours. How many hours would B need to complete the entire job alone?
20. At noon, a clock is correct. After that, its minute hand moves at the normal rate, but its hour hand moves 20% faster than normal. When the clock face first appears to show the hands exactly opposite each other, how many real minutes have passed since noon?

Part C. Beyond the Chapter

These questions are optional. They are not mainly about getting a numerical answer. Their purpose is to practice asking questions, deriving formulas, and summarizing ideas.

21. Derive the general section formula for a point x that divides the interval from a to b in the ratio $p : q$. Then explain in words why the coefficient of b is p , not q .
22. A slower traveler moves at speed v and has a head start of h hours. A faster traveler moves at speed $u > v$. Derive the catch-up formula

$$t = \frac{vh}{u - v}.$$

Then describe what happens to t as u gets closer and closer to v .

23. Derive a formula for the combined time of two workers who can complete a job separately in a and b hours. Your final formula should contain no fractions inside fractions.
24. Write a five-sentence summary of this chapter without using any of the numbers appearing in the examples. Your summary should explain at least one connection among ratios, motion, work, and clocks. Then create one new word problem whose solution uses two ideas from the chapter.

Answer Key

- | | |
|----------------------|---------------------|
| 1. 12, 18 | 2. 10 |
| 3. 9 oz | 4. 1 hour |
| 5. $7/2$ hours | 6. $10/3$ hours |
| 7. 6 hours | 8. 18 hours |
| 9. 10 degrees | 10. $6/5$ lb |
| 11. 43 | 12. 25 oz |
| 13. 150 seconds | 14. 40 seconds |
| 15. 50 mph | 16. 4 hours |
| 17. $360/11$ minutes | 18. 48 mph |
| 19. 12 hours | 20. $100/3$ minutes |

For Problem 21,

$$x = \frac{qa + pb}{p + q}.$$

For Problem 22,

$$t = \frac{vh}{u - v}.$$

For Problem 23, if the combined time is T , then

$$\frac{1}{T} = \frac{1}{a} + \frac{1}{b}, \quad T = \frac{ab}{a + b}.$$

Problem 24 is open-ended; a strong response should identify relationships, rates, units, and the habit of generalizing a solved problem.